TULJARAM PAVAN SINGH

Full-Stack MERN Developer • React & TypeScript Specialist • AI/ML Engineer

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PROFESSIONAL SUMMARY

Results-driven Full-Stack MERN Developer with 1+ year of hands-on production experience building and deploying scalable, high-performance web applications. Proficient in React.js, TypeScript, Node.js, Express.js, and MongoDB, with strong expertise in RESTful API integration, responsive UI development, and component-driven architecture using Tailwind CSS. Delivered production-ready features across gaming, 3D real-estate, and AI-powered SaaS platforms. Supplementary experience in Python-based Machine Learning, Deep Learning (Computer Vision), and OpenAI API integration. Actively seeking Full-Stack Developer, React Developer, MERN Stack Developer, or AI/ML Engineer roles.

TECHNICAL SKILLS

Languages	JavaScript (ES6+), TypeScript, Python, HTML5, CSS3, SQL
Frontend	React.js, Next.js, Redux, Tailwind CSS, Responsive Design, Accessibility (WCAG)
Backend & APIs	Node.js, Express.js, REST APIs, Socket.IO, JWT Authentication, Rate Limiting (Arcjet)
Databases	MongoDB, MongoDB Atlas, MySQL
AI / ML	Machine Learning, Deep Learning, Computer Vision, OpenCV, OpenAI API, Streamlit
Data Science	Python, Pandas, NumPy, Data Visualization, SQL
DevOps & Tools	Git, GitHub, Vercel, Postman, Figma, VS Code, Agile/Scrum
Auth & Security	JWT, OAuth, Clerk Authentication

PROFESSIONAL EXPERIENCE

Full-Stack Developer Intern | Ctruh
Jan 2025 – Dec 2025 • Hyderabad (Remote)

- Built and shipped 20+ reusable, modular React components with TypeScript across 5+ production platforms (gaming, 3D real-estate, enterprise CMS), reducing feature delivery time by ~30%.
- Delivered pixel-perfect, responsive UIs achieving sub-2-second load times; optimized rendering performance using lazy loading, code splitting, and memoization.
- Integrated frontend components with RESTful backend APIs and CMS workflows, ensuring seamless data flow and reliable end-to-end functionality.
- Implemented WCAG-compliant accessibility features — voice-over support, semantic HTML, ARIA attributes — improving inclusive UX across platforms.
- Executed end-to-end testing, UI validation, and bug triage across 5+ production releases in an Agile/Scrum team of frontend, backend, and design stakeholders.
- Collaborated closely with UI/UX designers and backend engineers to translate Figma designs into production-ready code with high fidelity.

KEY PROJECTS

JourneyBot.AI – AI-Powered Trip Planner *Sep 2025 – Nov 2025 Next.js · TypeScript · Node.js · MongoDB Atlas · OpenAI API · Google Places API · Mapbox · Clerk · Arcjet · Tailwind CSS · Vercel*

- Architected and launched a full-stack AI SaaS platform that generates personalized multi-day travel itineraries with hotel recommendations using OpenAI API.
- Implemented secure JWT-based authentication via Clerk, API rate-limiting with Arcjet, and interactive geolocation maps using Mapbox GL JS.
- Deployed on Vercel with MongoDB Atlas backend; integrated Google Places API for real-time location search and enriched itinerary data.

Gaming Platform UI
React.js · TypeScript · Tailwind CSS · Framer Motion

- Engineered a visually rich, animated homepage with interactive components and complex responsive layouts, achieving consistent performance across all device sizes.
- Optimized animation rendering to maintain 60fps performance; reduced Largest Contentful Paint (LCP) through asset optimization and lazy loading.

3D Real-Estate Platform UI
React.js · Tailwind CSS · Three.js

- Converted high-fidelity Figma designs into pixel-perfect, production-ready React components for a 3D interactive real-estate showcase platform.
- Implemented dynamic layouts and smooth scroll-based animations, improving user engagement and time-on-site metrics.

Virtual Jewellery Try-On Module
React.js · Tailwind CSS · Component Design System

- Designed a reusable component library (card layouts, image holders, multi-variant displays) for a virtual try-on product, maintaining UI consistency across the platform.
- Collaborated with the core AR/try-on logic team to integrate UI with underlying computer vision processing pipeline.

Deep Learning Accident Detection System
Jan 2024 – Mar 2024 *Python · OpenCV · TensorFlow · Streamlit · Twilio API*

- Developed a real-time deep learning system to detect road accidents from video feeds with ~90% accuracy in controlled test environments.
- Built automated SMS alert notification system using Twilio API; deployed interactive demo via Streamlit for stakeholder presentation.

EDUCATION

B.Tech in Information Technology
2020 – 2024

Vidya Jyothi Institute of Technology, Hyderabad | CGPA: 6.7/10

CERTIFICATIONS

- Crash Course on Python — Google
- Databases and SQL for Data Science with Python — IBM
- Python for Data Science — IBM
- Java Programming — Infosys Springboard

LANGUAGES

English (Professional Fluency) • **Hindi** (Native) • **Telugu** (Intermediate)